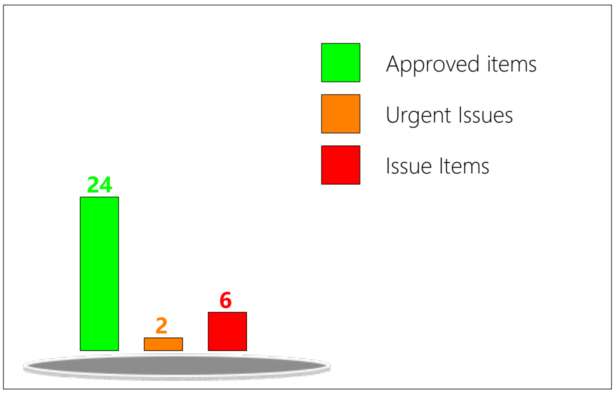
HQDFM Design for Manufacture(DFM) Report

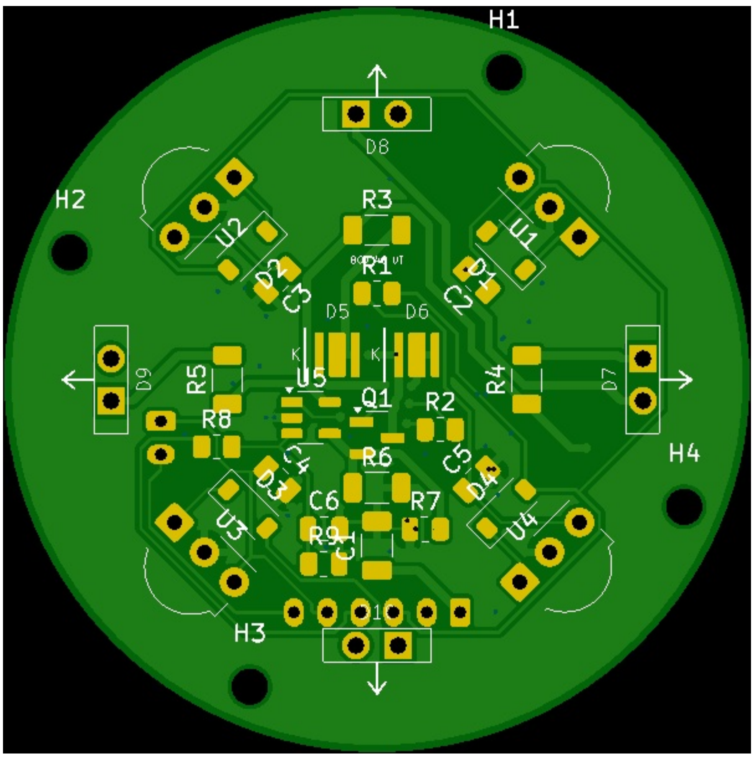
File name: 2026-06-21

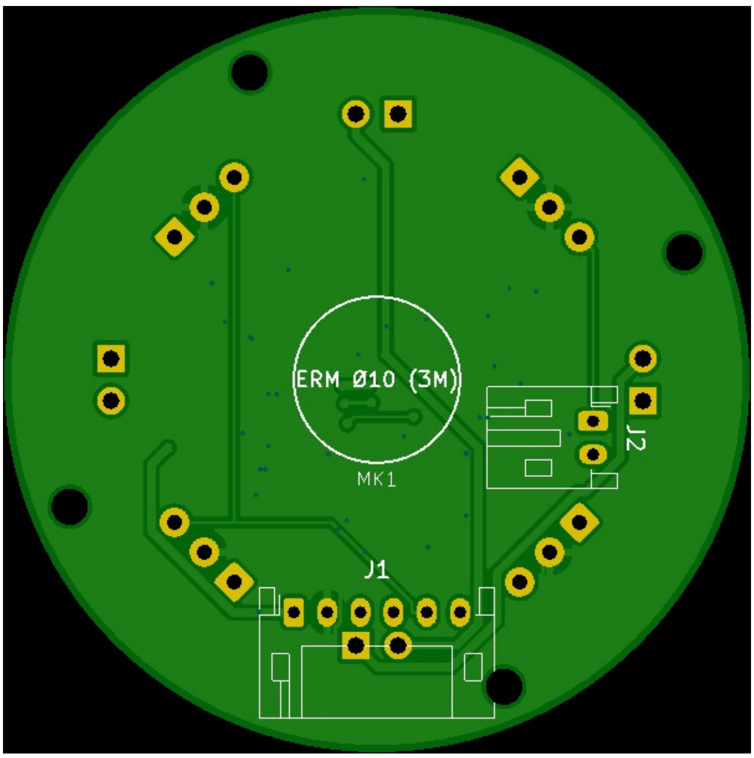
Time: 2026-06-21 Layer count:2 PCB Thickness: 1.60 Quantity: 5 mm



Basic Board Specs	Trace Width/Spacing	8.00/8.00mil
	Milling Density	72.9977m/m²
	Surface Finish Area	15.39%
	Test Point Count	80

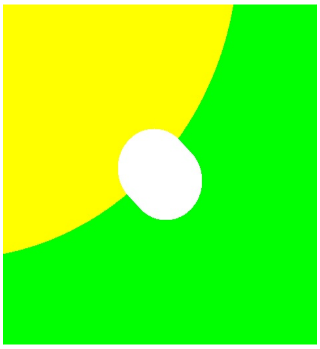
The file size is small, which can affect the surface mount assembly process. It is recommended to have a size larger than 7*7cm. You can optimize the size by adding a process edge or increasing the panelization.

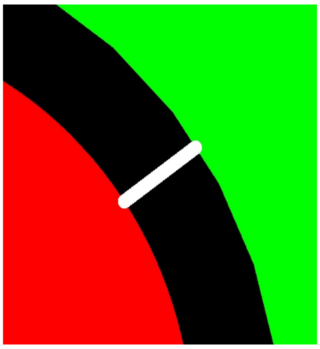
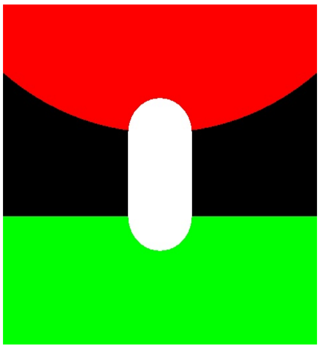


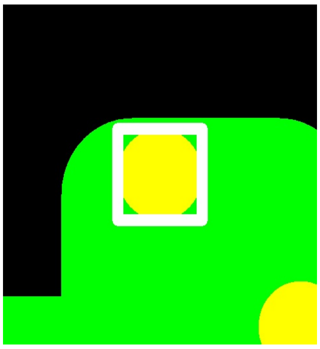
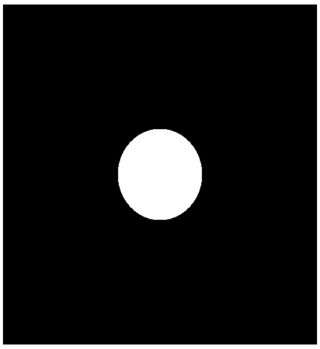


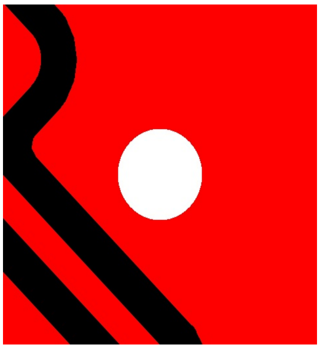
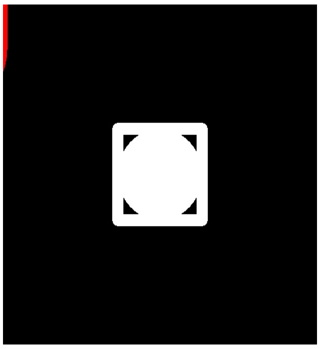
Type	Category	No. of Checks	Result
PCB Trace Analysis	Open/Shorts (IPC)	1	Fail
	Signal Integrity	4	Fail 5
	Smallest Trace Width	1	Pass 2
	Smallest Trace Spacing	3	Pass 178
	SMD Pad Spacing	1	Pass
	Pad Size	3	Pass 5
	Hatched Copper Pour	2	Pass
	Annular Ring Size	2	Pass 4 , Fail 4
	Drill to Copper	5	Pass 50 , Fail 12
	Copper-to-Board Edge	2	Pass 4
	Holes on SMD Pads	4	Fail 3
PCB Drilling Analysis	Drill Diameter	8	Pass 12
	Drill Spacing	4	Pass
	Drill to Board Edge	4	Pass
	Drill Hole Density	1	Pass
	Special Drill Holes	2	Pass
	Drill Hole Errors	3	Fail 52
PCB Solder Mask Analysis	Solder Mask Dam	2	Pass 4
	Missing SMask Opening	1	Fail 60
	Solder Paste Area	1	Pass
PCB Silk Analysis	Silkscreen Spacing	1	Pass 12 , Fail 18

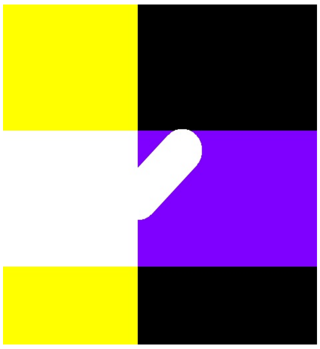
PCBA Component Analysis	Component Spacing	1	Fail
	Comp.-to-Board-Edge	3	Fail
	Component Silkscreen Spacing	0	Fail
	Pad Count Mismatch	4	Fail
	Designator Length	0	Fail
	Double-sided Components	1	Pass
	Mid Mount Comp	1	Pass
	Comp Height Analysis	2	Pass
PCBA Pin Analysis	SMT Pad Analysis	8	Fail
	Through-hole Pins	10	Fail
PCBA Pad Analysis	Chip Pads	60	Fail
PCBA Fiducial Analysis	Fiducial Count	1	Fail
	Fiducial Analysis	4	Pass

ID	Check	Limits	Value	Issue	Image	Position	Qty	Level
1	Signal Integrity_Acute Angle Traces	-,-,-	Error(s) detected	There is an acute angle connection mode in your design,which will affect the signal integrity of the product.it is recommended to adjust the acute angle posintion to arc or obtuse angle connection mode		147.45,-133.72	1	Warning
2	Annular Ring Size_PTH Annular Ring	6,7,8	0.01 mm	PTH Pad Rings0.34 mil in size were detected in your design. This could increase the risk of tangency or breakouts, which decrease electrical reliability and yield, and affect the reliability of the boards. It is recommended that PTH Pad Rings have a width of at least 6 mil.		155.43,-113.48	7	Risk

3	Drill to Copper_NPTH-to-Copper	8,10,12	0.05 mm	NPTH to copper spacing of 9.86mil was detected in your design. This could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 12 mil.		143.35,-137.78	8	Risk
4	Drill to Copper_PTH-to-Trace [Outer]	10,12,14	0.05 mm	For most factories, the PTH to trace clearance requirement in outer copper layers is at least 10 mil. Failure to meet the factory's requirement could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. PTH to trace spacing (outer layer) of 1.83mil was detected in your design. It is recommended to increase the spacing to at least 12 mil.		138.42,-123.45	53	Risk

5	Holes on SMD Pads_PTH on SMD Pad	0.1,0.05,0	100.00 %	Holes on surface mount pads were detected in your design. During SMT assembly, solder could leak into the hole and pull solder away from the SMD contact, which could decrease manufacturing efficiency and yield, and affects the reliability of the boards. Please check and separate the holes from the pads if possible.		151.82,-128.47	3	Risk
6	Drill Hole Errors_Stray Drills	-,-,-	Error(s) detected	Stray drill holes were detected in your design. These are holes that do not have corresponding pads or solder mask openings. The stray drill holes may be a result of problems with the drill hole files which cause the holes to be misaligned or produce additional drill hits which could cut into traces or pads. Please check the export configuration and export the drill files again if necessary.		138.42,-123.27	28	Warning

7	Drill Hole Errors_Missing Drills	-,-,-	Error(s) detected	Missing drill holes were detected in your design. This could result in circuit opens and prevent through hole parts from being inserted correctly. The missing drill holes may be due to a problem with the drill files, please check the export configuration and export the drill files again if necessary.		143.43,-108.57	24	Warning
8	Missing SMask Opening_Missing SMask Opening	-,-,-	Error(s) detected	Pads with no corresponding solder mask opening in the solder mask layer were detected in your design. Please check and modify the files if necessary. If it is intentional, please make a note of it to the manufacturer.		138.42,-123.27	33	Risk

9	Silkscreen Spacing _Solder Mask-to-Silkscreen	4,5,6	Error(s) detected	For most factories, the minimum silkscreen to solder mask spacing requirement is at least 8 mil. Failure to meet the factory's requirements could result in part of the silkscreen being removed or being printed directly on the pads, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. Silkscreen to solder mask spacing of 0 mil were detected in your design. It is recommended to increase the spacing to at least 12 mil.		152.14,-135.98	2	Risk